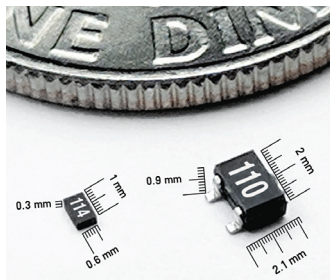




and 2mm x 2.1mm x 0.9mm (PCC110), and have up to 75 % conversion efficiency while harvesting energy from all



modulation types, enabling maximum power transfer and range. Due to their small size, they have a low cost of implementation and are RoHS compliant.

Powercast Corporation
www.powercastco.com

DSP and ML coupling chips

Tyr, a family of three different chips—called Tyr1, Tyr2, and Tyr3—uses a fully programmable and scalable architecture that tightly couples digital signal processing (DSP) cores with machine learning



(ML) accelerators necessary to design L3 through L5 autonomous driving vehicles. Delivering between 258-trillion

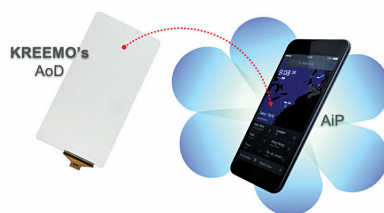
and 1,032-trillion operations per second and consuming as little as 10 watts, Tyr allows users to implement autonomous driving functions previously not commercially viable. The Tyr companion chip is algorithm and host processor agnostic and can be integrated into new or existing environments without the need to redesign the entire system.

VSORA
www.vsora.com

Transparent built-in display antenna

The world's first transparent antenna on display (AOD) is characterised by a smooth 360-degree high-capacity data transmission and reception. By combining an antenna and a display into one, more space can be utilised. The built-in display antenna can be mounted on

high-definition screens and glasses in organic light emitting diodes (OLEDs) or liquid crystal displays (LCDs) of

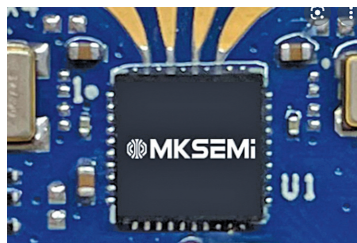


smartphones, augmented reality (AR), and game devices. The antenna technology is expected to become a metaverse antenna platform in a rapidly-growing global market.

KREMO
www.kreemo.io

Low-power, integrated SoC

The MK8000 system-on-chip (SoC) is the world's lowest power, highest integrated chip solution that can empower IoT devices with the smallest batteries and size for high-precision sensing. MK8000 also enables OEMs and ODMs to rapidly design, integrate, and deploy



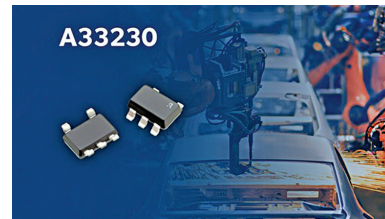
complete location and direction-finding solutions. With a high level of integration and computing power, the MK8000 SoC allows customers to implement UWB ranging products with only a handful of external components, ensuring the smallest possible footprint and lowest power consumption. Its low-power UWB technology significantly extends the battery life of electronics and enables even the smallest wireless devices to have high precision ranging and location capability.

Mauna Kea Semiconductors
www.mk-semi.com

Tiny Hall-effect position sensor

The new A33230 3D sine/cosine Hall-effect position sensor IC is the small-

est 3D sine/cosine sensor currently available in the market. It offers system designers a cost-effective solution for



automotive and industrial applications with a quick time to market. The A33230 contains two discrete analogue signal paths for superior high-speed performance, and the tiny SOT23-W package size makes it ideal for situations where PCB space is at a premium. The IC can also be used as an angle sensor, provided the system includes an electronic control unit (ECU) capable of performing CORDIC calculations on the two outputs.

Allegro MicroSystems
www.allegromicro.com

Mini CPC capacitive sensors

Two unique, miniature carbon-nano-tube paper composite (CPC) capacitive sensors, SomaControl and SomaSense, bring a better sense to the digital world by creating a new genre of gesture based digital interfaces, wellness monitoring, and safety applications. The SomaControl gesture monitor enables everyday tasks at home or integrated



into gaming devices for a more immersive experience. It also allows users to interact with and control a digital device using hand movements with no contact. The SomaSense 3D floor mat is a flexible sensing mat that observes, monitors, and reports on human wellness factors, including presence, gait, and foot pressure, with applications in health-tech and wellness, assisting individuals with balance, movement, and other challenges.

Somalytics
www.somalytics.com